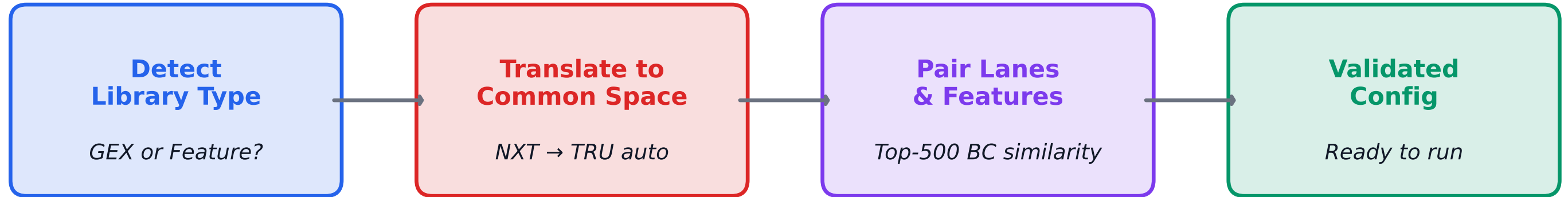


Perturb-seq FASTQ Pre-Flight

Automated library detection, namespace translation, and lane pairing



- Reads barcode structure to classify each FASTQ as Expression or Feature
- Auto-translates Feature barcodes to GEX namespace (NXT → TRU)
- Computes top-500 barcode similarity to join lanes and features to GEX
- Catches and corrects user mixups, mislabels, and omissions
- Tested on UCSF perturb-seq set — runs in seconds